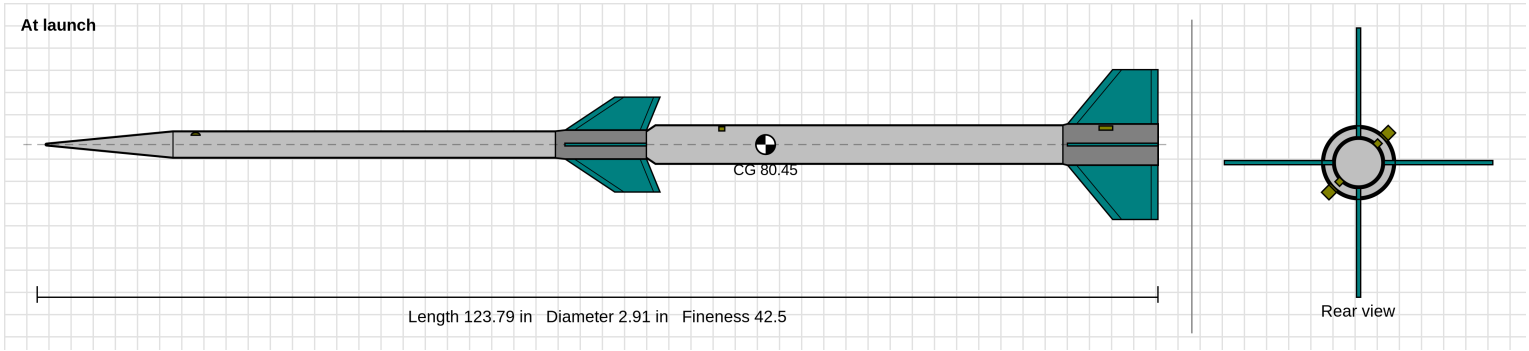
PANDORA - AMA_CONFIG_15 - Results



PANDORA - AMA_CONFIG_15
Nose radius for heating: 0.1 in

	nominal	mean	std	-2 sigma	+2 sigma	75% low	75% high	lowest	highest	
Rail exit speed	115.3	115.37	1.282	112.8	117.93	113.88	116.85	112.12	117.87	ft/s
Rail exit AoA	3.7515	3.7207	2.629	-1.5373	8.9788	0.67274	6.7687	3.8564	3.6695	deg
Forecast gust	7.874	12.542	10.714	-8.8856	33.969	0.12051	24.963	7.874	7.874	ft/s
Rail exit AoA in gust	3.9068	6.1429	5.2206	-4.2984	16.584	0.090244	12.196	4.016	3.8214	deg
Apogee	4.0891e+05	4.0584e+05	21688	3.6246e+05	4.4922e+05	3.807e+05	4.3099e+05	3.3829e+05	4.7082e+05	ft
Time to apogee	173.03	172.24	4.5207	163.19	181.28	166.99	177.48	157.17	185.34	s
Max speed	6080.6	6077	122.46	5832.1	6321.9	5935	6219	5739	6423.8	ft/s
Max Mach	6.228	6.2232	0.12904	5.9651	6.4813	6.0736	6.3728	5.8503	6.5959	
Max dynamic pressure	8861.4	8972.1	334.35	8303.4	9640.8	8584.5	9359.7	8932.3	9147.1	psf
Peak heat flux	951.15	952.6	53.098	846.4	1058.8	891.04	1014.2	854.77	1078.6	BTU/ft²/s
Heat load	6535.4	6540.3	275.5	5989.4	7091.3	6220.9	6859.7	5770.5	7278.1	BTU/ft²
Tilt at ignition	3.2311	4.1755	2.3599	-0.54423	8.8952	1.4395	6.9115	3.7895	2.432	deg
Altitude at ignition	31485	31448	890.78	29667	33230	30415	32481	29142	32879	ft
Flight time	567.97	566.74	9.6752	547.39	586.09	555.52	577.96	537.49	597.13	s
Landing east	-2.96	-11.017	9.0626	-29.142	7.1081	-21.524	-0.51022	-1.9899	2.7756	mi
Landing north	-9.6118	-10.201	9.6575	-29.516	9.114	-21.398	0.99568	-8.497	-9.2432	mi
Impact speed	36.042	36.414	1.1073	34.199	38.628	35.13	37.698	36.065	35.459	ft/s
Launch mass	71.337	lb, CG 80.45 in from the nose tip								
Min static margin	-8.5437	cal (while loaded)								
Max angle of attack	0.40602	deg (while loaded)								

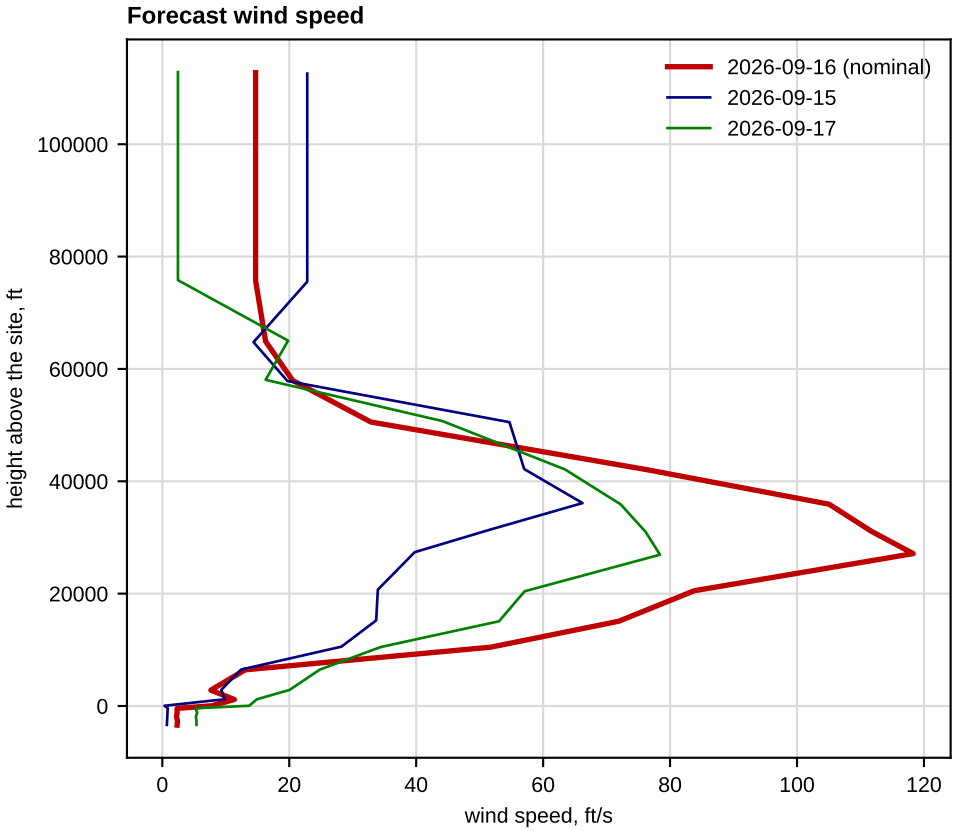
Sustainer ignited: yes
  Monte Carlo: 128 of 128 ignited
Monte Carlo runs: 128
  -2 sigma and +2 sigma are the mean less and plus two standard deviations.
  75% low and high: the prediction interval, where 75% of flights fall: mean -/+ z s sqrt(1 + 1/N),
  z = 1.150, N = 128 runs, assuming a normal spread.
  lowest and highest are the runs with the lowest and highest apogee (runs #50 and #117).
Launch cylinder, 17 nmi: the nominal flight lands inside, 80 of 128 Monte Carlo runs land inside

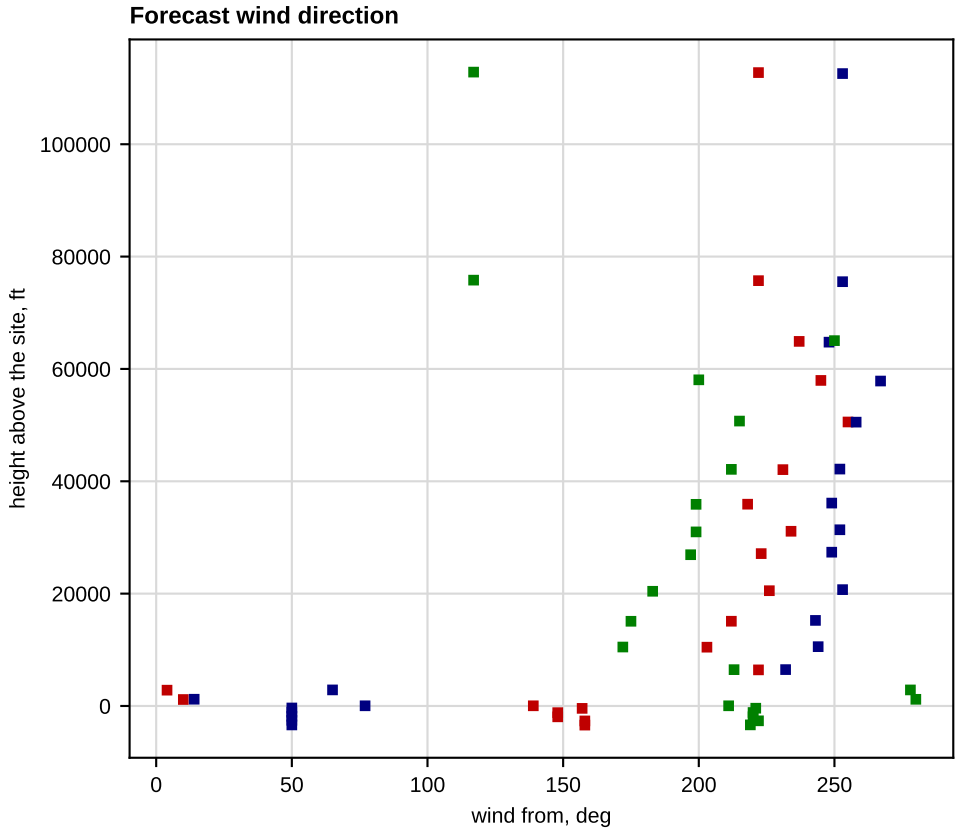
Events (nominal flight)
  3.43 s Booster burnout, separation
 18.43 s Sustainer ignition
 21.52 s Sustainer burnout
173.03 s Apogee
173.03 s Apogee chute
513.80 s Main chute

PANDORA - AMA_CONFIG_15 - Weather and Monte Carlo inputs

```
Weather
  Open-Meteo forecast at 40.8707, -119.1069, 06:00 America/Los_Angeles
  day                   surface wind         gust at the launch hour   runs
  2026-09-16      2.3 ft/s from 158 deg    7.9 ft/s from 139 deg at 06:00     41   nominal flight
  2026-09-15      0.7 ft/s from  50 deg    1.0 ft/s from  77 deg at 06:00     39
  2026-09-17      5.4 ft/s from 219 deg   25.9 ft/s from 211 deg at 06:00     48
  Each day is a whole forecast wind profile; the gust is the 10 m forecast for the launch hour, used only for the rail exit AoA in gust.

Monte Carlo inputs: these 8 change from run to run, everything else is fixed
                            nominal     sigma      lowest      mean   highest   (as drawn)
  Booster impulse               100         2       94.14     100.1     104.6  %
  Sustainer impulse             100         2       94.84     99.74       104  %
  Booster dry mass              100         2       95.23     99.95     105.7  %
  Sustainer dry mass            100         2       95.54     99.89     105.1  %
  Coast before ignition          15       0.5       13.58     14.96     16.14  s
  Launch angle                   90       0.5       88.37     89.59        90  deg
  Heading                         0       180      -177.3    -12.13     178.6  deg
  Forecast day            2026-09-16   one of the 3 days above, picked at random
  Random seed: different every time
```





PANDORA - AMA_CONFIG_15 - Inputs

```
[Case]
  Units               IN
  Analyze             STACK

[Nose]
  Shape               CONICAL
  Length              15
  PowerExponent       0.5
  TipRadius           0.1
  FlatTip             false

[Body]
  Diameter            2.91
  Length              52.29
  BoattailLength      0
  BoattailDiameter    3
  NozzleExitDiameter  2.5

[Fins]
  Count               4
  RootChord           9
  TipChord            5
  Span                3.65
  Sweep               5.5
  AftOffset           0
  Thickness           0.147
  Section             CHAMFER
  LEChamfer           1.019
  TEChamfer           0.7

[FinCan]
  Enabled             true
  Diameter            3.15
  Length              10.05
  AftOffset           0
  FrontBevel          1.09

[Booster]
  Enabled             true
  Diameter            4.25
  Length              56.5
  Insertion           0
  TransitionLength    1
  NoseCone            false
  NoseShape           OGIVE
  NoseLength          8
  BoattailLength      0
  BoattailDiameter    0.01
  NozzleExitDiameter  3.5

[BoosterFins]
  Count               4
  RootChord           10
  TipChord            5
  Span                6
  Sweep               5
  AftOffset           0
  Thickness           0.187
  Section             CHAMFER
  LEChamfer           1
  TEChamfer           0.75

[BoosterFinCan]
  Enabled             true
  Diameter            4.5
  Length              10.5
  AftOffset           0
  FrontBevel          0.5

[Protuberance1]
  Type                FAIRING
  Stage               SUSTAINER
  X                   17
  Count               2
  Length              1
  Width               0.4
  Height              0.4
  Standoff            0

[Protuberance2]
  Type                BUTTON
  Stage               BOOSTER
  X                   8
  Count               2
  Length              1
  Width               0.65
  Height              0.65
  Standoff            0

[Protuberance3]
  Type                LUG
  Stage               BOOSTER
  X                   50
  Count               2
  Length              1.5
  Width               0.65
  Height              0.65
  Standoff            0

[Flight]
  Altitude            4000
  ClipFinsToShock     false
  TurbulentBoundaryLayertrue

[Motors]
  Sustainer           N3364.eng
  Booster             O8220.eng

[Site]
  Weather             OPENMETEO
  Latitude            40.8707
  Longitude           -119.107
  Elevation           3701
  Date                2026-09-16
  Time                06:00
  Timezone            America/Los_Angeles
  WindSpeed           0
  WindDirection       0
  CylinderRadius      17

[Launch]
  RailLength          15
  Angle               90
  Heading             0
  Coast               15
  MaxTilt             10
  MinIgnitionAltitude 0
  MaxTime             1000

[Recovery]
  ApogeeChuteDiameter 8
  ApogeeChuteCd       0.8
  MainChuteDiameter   28
  MainChuteCd         1.8
  MainChuteAltitude   2000

[Thermal]
  IncludeDescent      false

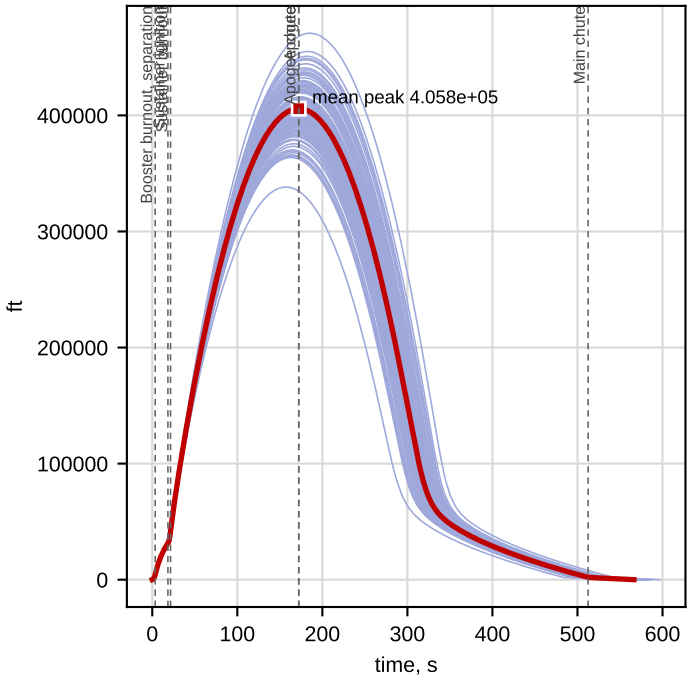
[DrySustainer]
  Mass                10.092
  CG                  41.9
  Inertia             4500
  MotorOffset         2.6

[DryStack]
  Mass                34.387
  CG                  85
  Inertia             30212.9
  MotorOffset         3.5

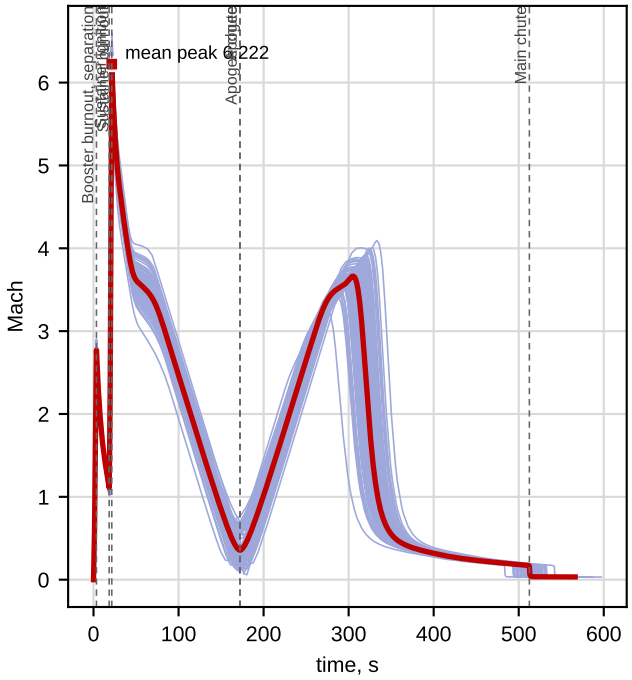
[MonteCarlo]
  Runs                128
  Seed                0
  Prediction          75
  AngleStd            0.5
  HeadingStd          180
  CoastStd            0.5
  BoosterImpulseStd   2
  SustainerImpulseStd 2
  BoosterMassStd      2
  SustainerMassStd    2
  WindSpeedStd        0
  WindDirectionStd    0
  WindDays            1
  MaxDayWind          10
  SkipDays
```

AMA_CONFIG_15 - Flight, mean of 128 Monte Carlo runs

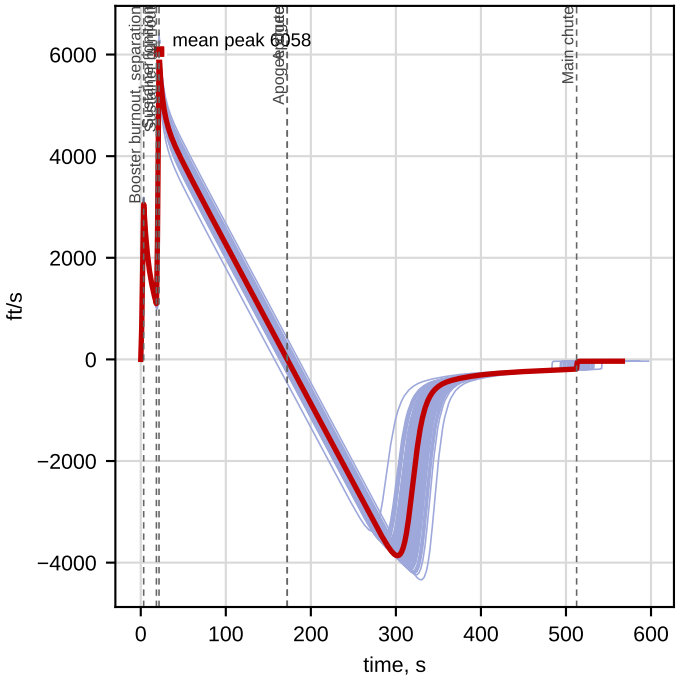
Altitude



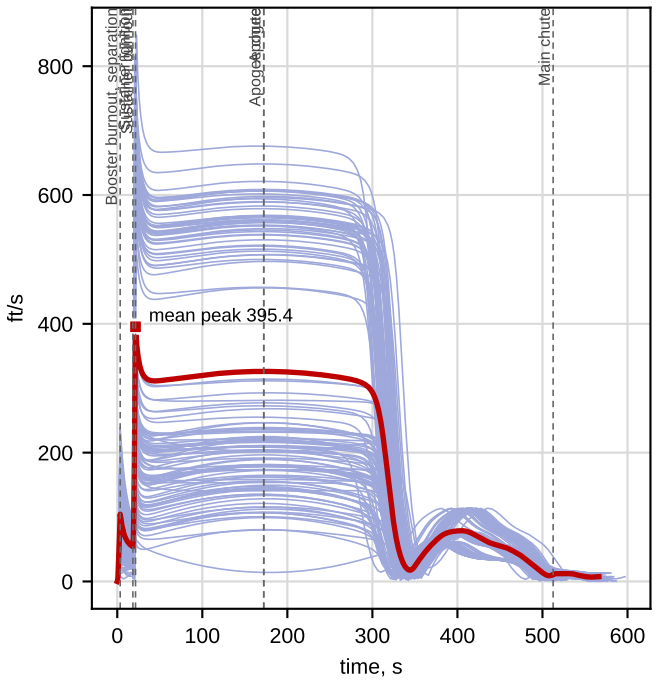
Mach number



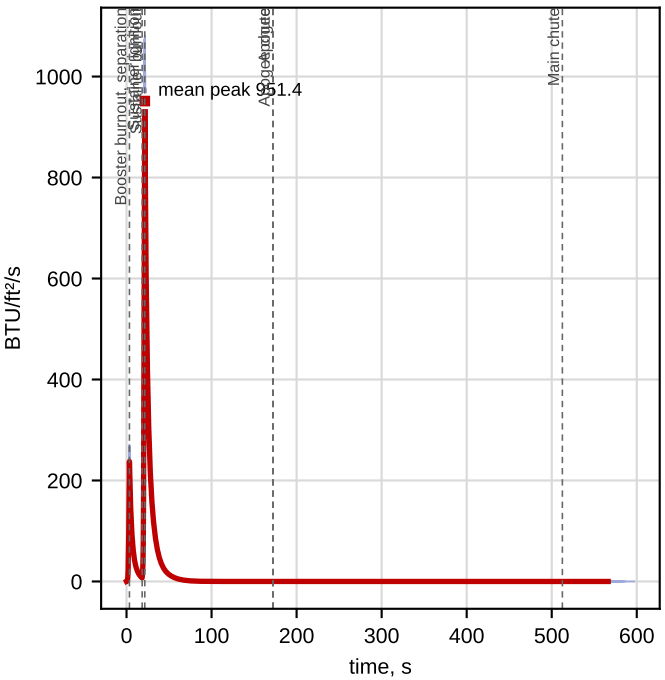
Vertical velocity



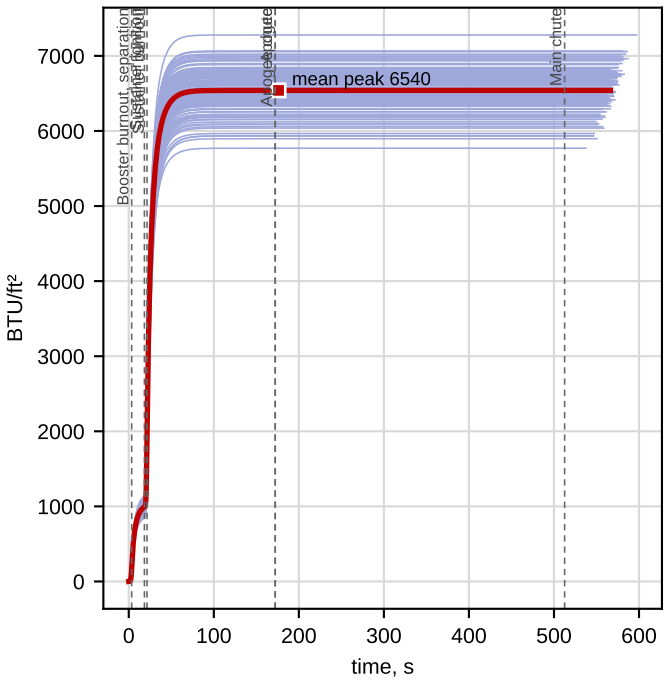
Horizontal velocity



Stagnation heat flux

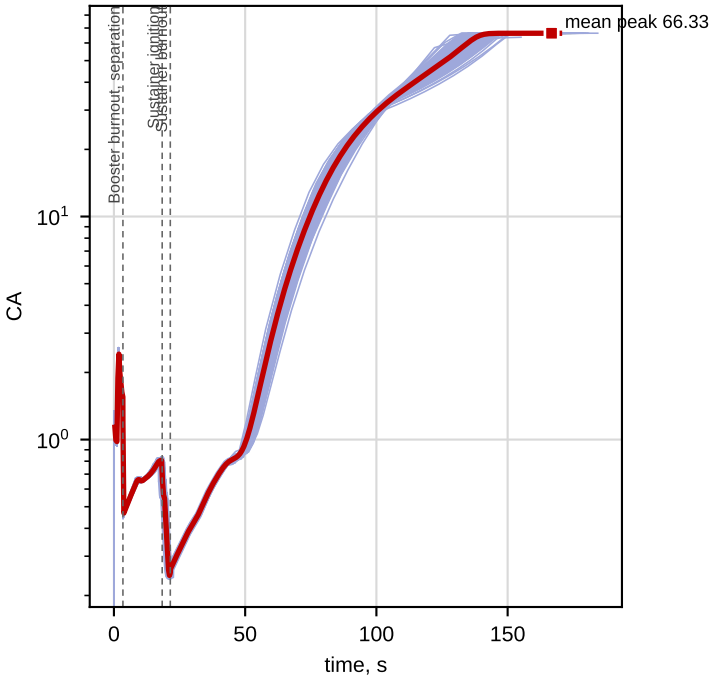


Integrated heat load

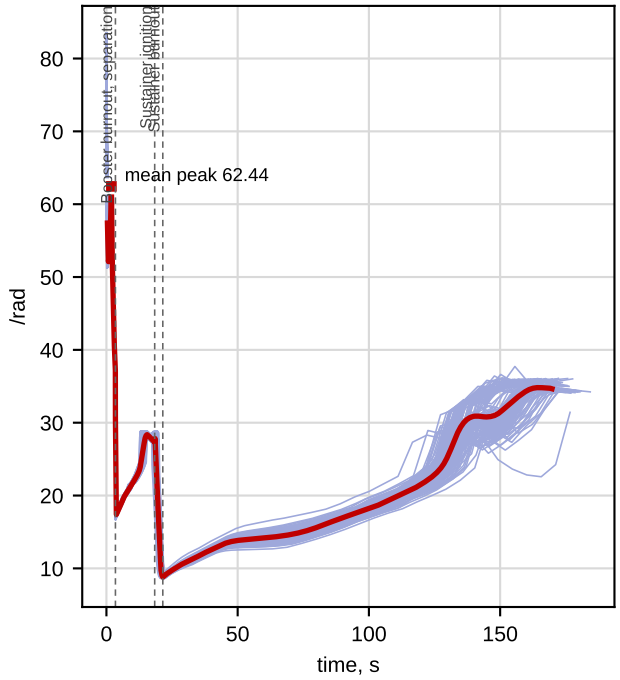


AMA_CONFIG_15 - DATCOM aerodynamics along the flight (launch to apogee, Mach 0.1 and up)

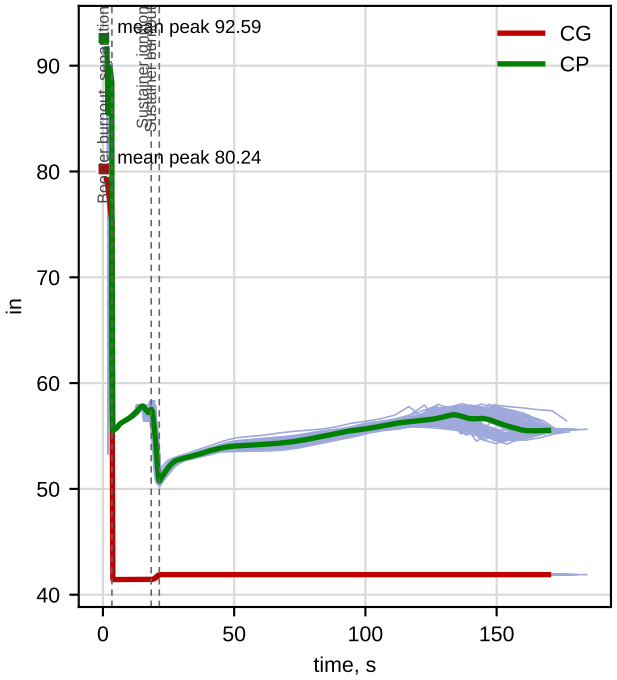
Axial force coefficient CA (log scale)



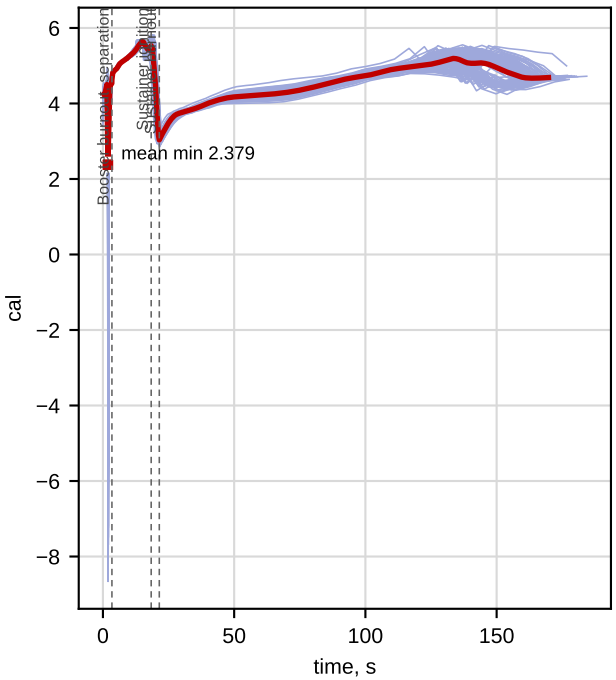
Normal force slope CNα



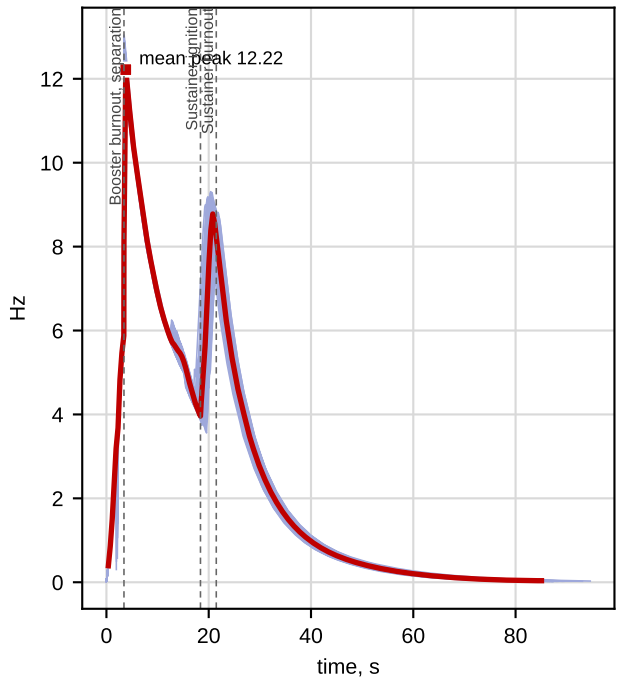
CG and centre of pressure



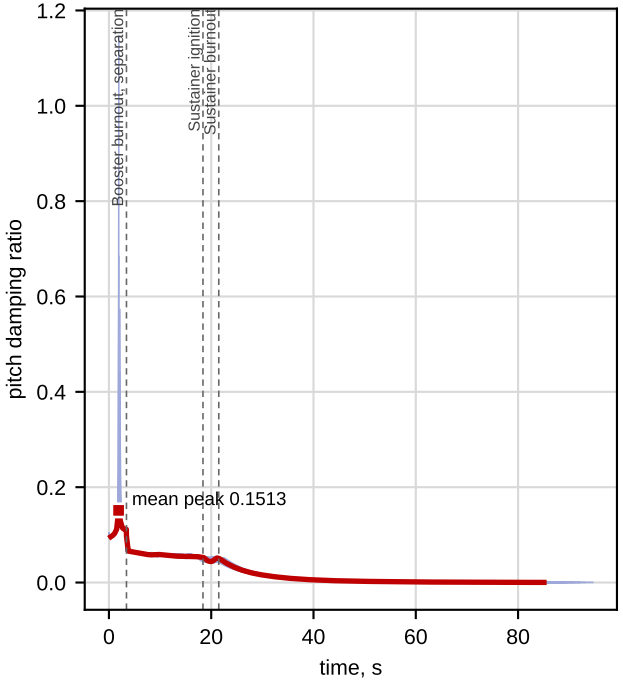
Static margin



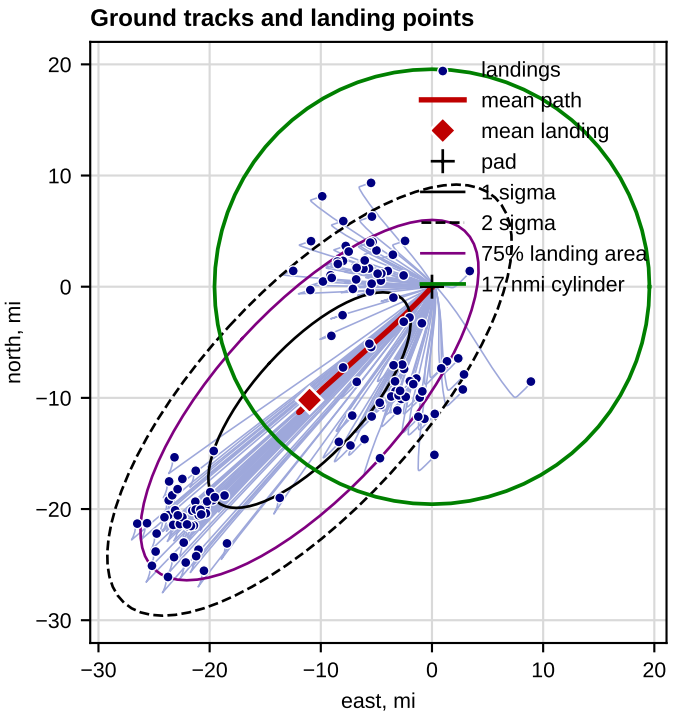
Pitch natural frequency

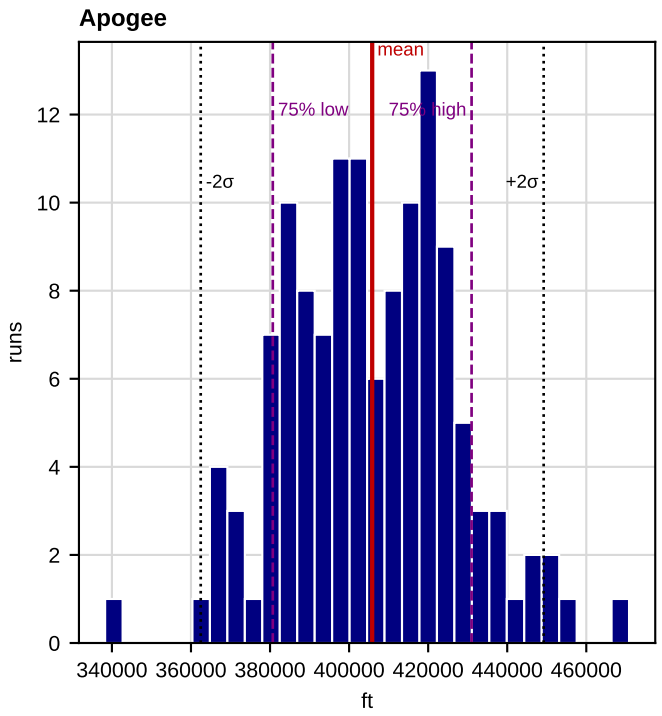


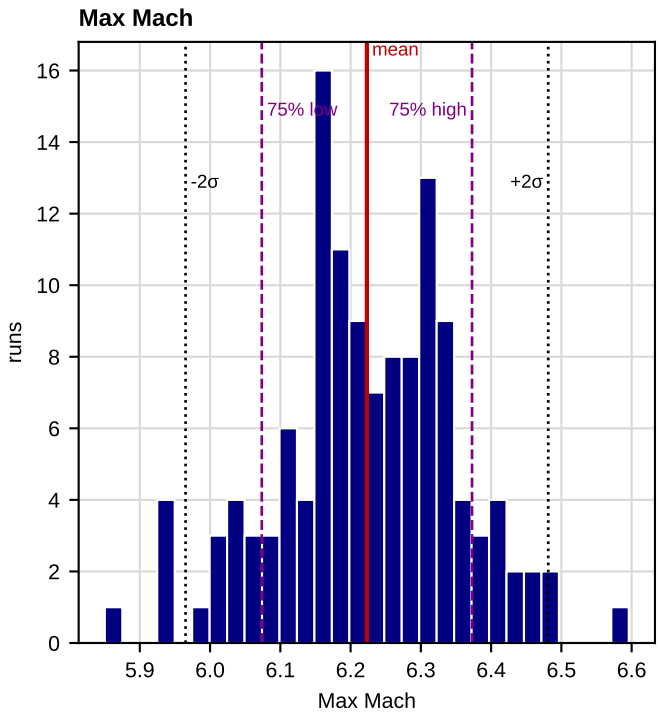
Pitch damping ratio

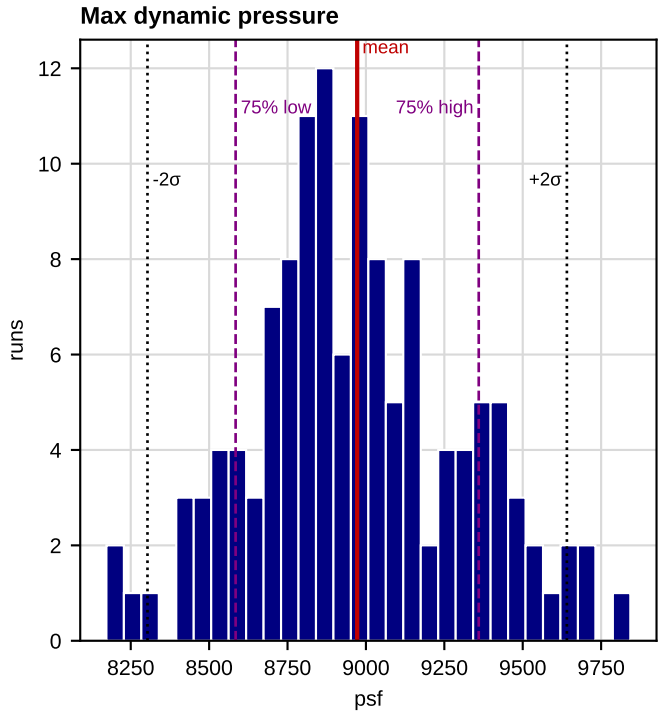


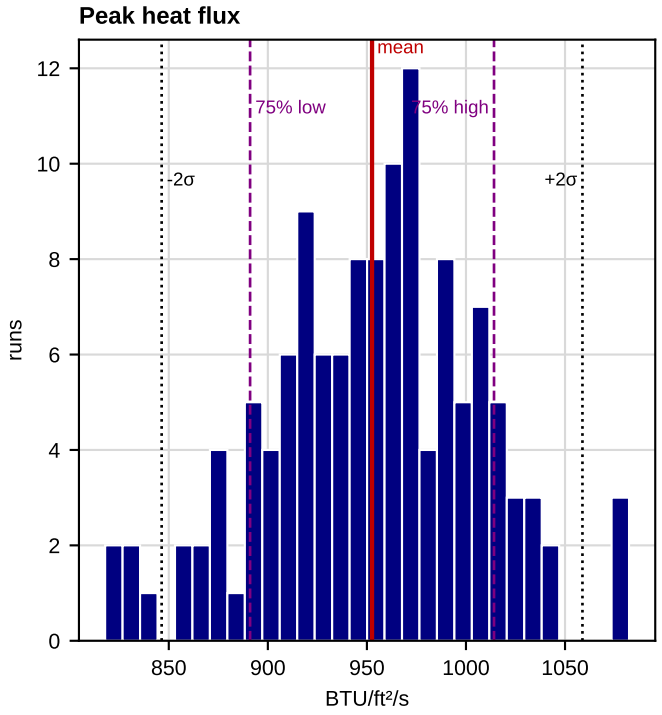
AMA_CONFIG_15 - Monte Carlo, 128 runs

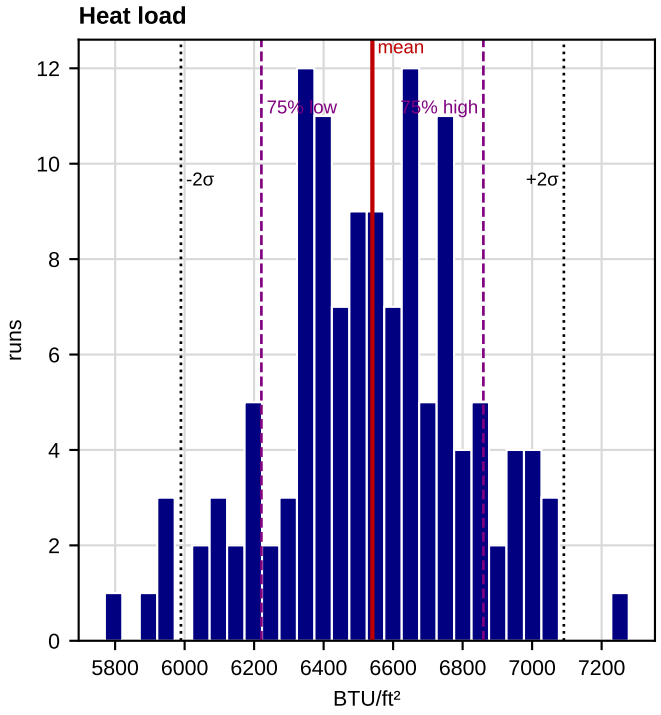






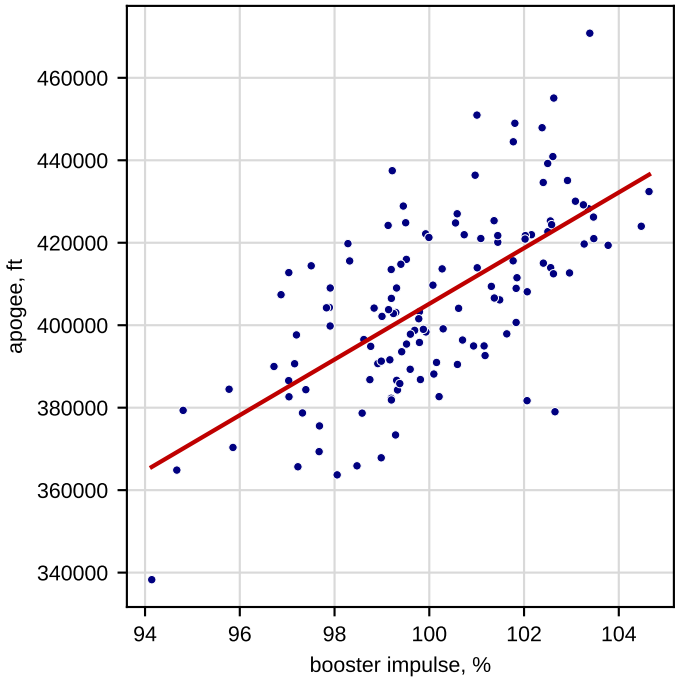




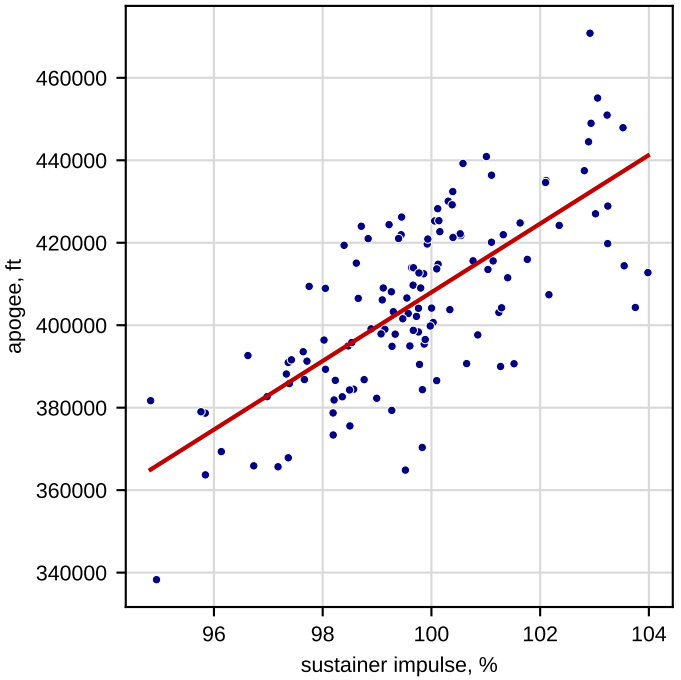


AMA_CONFIG_15 - Sensitivity: what each varied input did to apogee, 128 runs

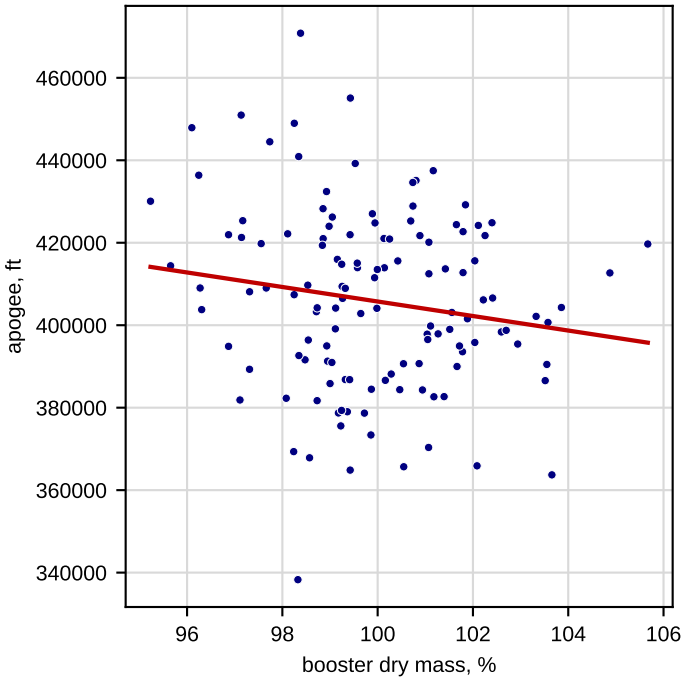
Booster impulse: +6753 ft per %



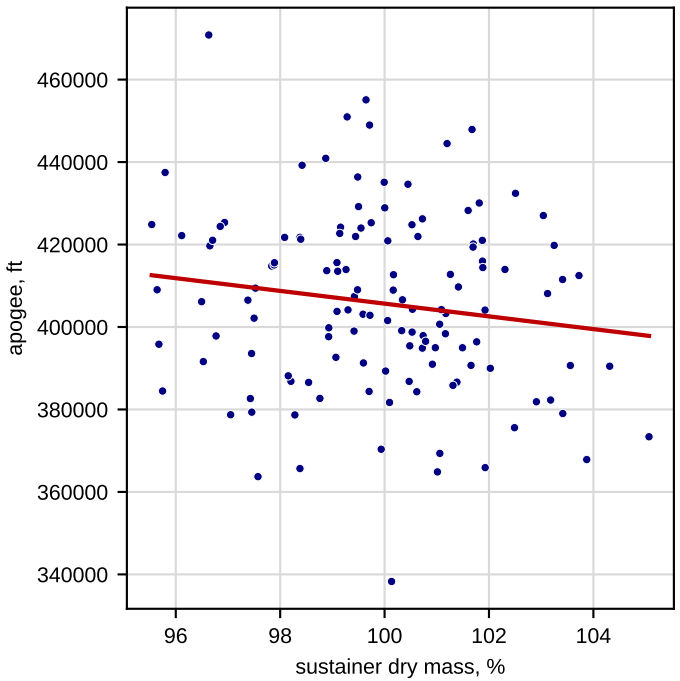
Sustainer impulse: +8324 ft per %



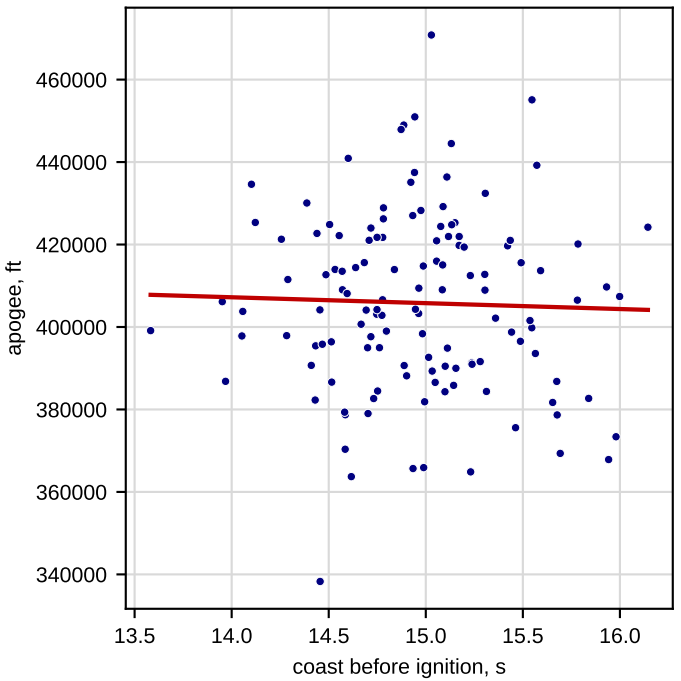
Booster dry mass: -1761 ft per %



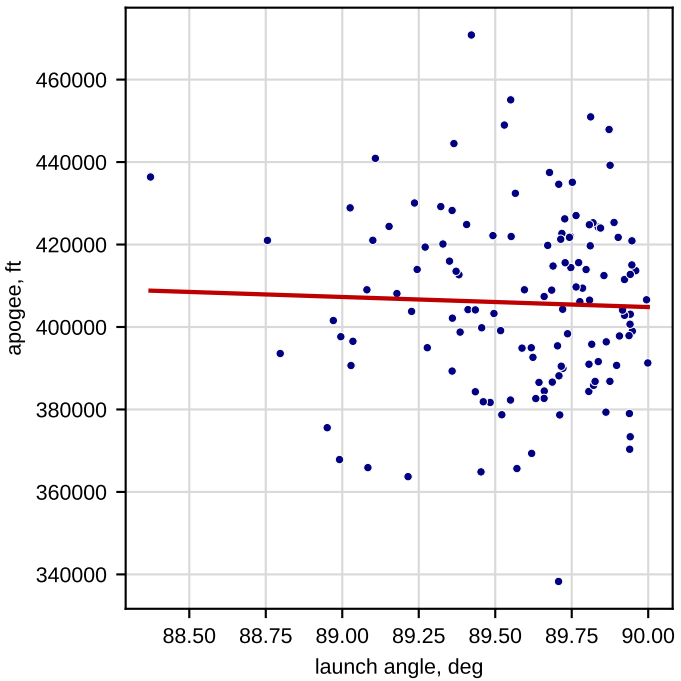
Sustainer dry mass: -1544 ft per %



Coast before ignition: -1426 ft per s



Launch angle: -2447 ft per deg



AMA_CONFIG_15 - Ground track, 128 Monte Carlo runs

